pdf-to-markdown

Release notes, version 2.4

11 October 2026

Summary

`pdf-to-markdown` is a Claude skill that converts PDFs of physics papers, notes and textbooks into files for a Claude Project: a text-only Markdown file, named exactly like the original PDF, in which every equation is written in LaTeX, and one companion PDF holding the document's figures. Every formula is transcribed from page images rather than from the PDF's text layer, then checked mechanically and proofread against the original.

Section 2 analyses what this gains over uploading the original PDF. In short: about 20–30% less Project capacity per paper with all figures included, and up to 40–45% without them; exact equations instead of a garbled text layer or small page images; content that remains searchable when a large Project switches to retrieval; and, the largest effect, no image-reading work whenever a conversation refers to equations, which saves an estimated 20,000–47,000 tokens in an equation-heavy chat. A conversion costs about 35,000–45,000 tokens for an eight-page paper once, so it pays for itself within one or two such chats.

All figures in this note are estimates unless stated as measured. The evidence comes from born-digital documents; transcription has been tested on single-column documents only (Section 7).

Version history

Version	Change
1.0	Draft, never packaged: markitdown text plus re-transcription of "math-heavy" pages from 200 dpi images.
2.0	First release: math transcribed from strips cut between lines; `check.py` and `verify_eqs.py`; evaluation scripts.
2.1	Scorer rewritten: content-based alignment, tolerance for spelling-only variants, `--lenient`, exact-match counts.
2.2	Token efficiency: 170 dpi strips, no page-by-page reading of the text layer, per-page output joined by `assemble.py`, compact printout.
2.3	Figures: `figures.py` finds figures by caption and renders them; first delivered embedded in the Markdown.
2.4	Output for Projects: text-only Markdown named exactly like the PDF, plus `NAME_figures.pdf` packed adaptively (at least 150 dpi per figure); embedding removed. Within 2.4 the figure delivery moved from separate image files to a zip and then to the figures PDF, the only form a Project accepts as one file.

1 What the skill does

A conversion runs in six steps, all described in `SKILL.md`:

1. `prepare.py` cuts every page into 170 dpi strips at the gaps between lines and prints, per page, the strips with their positions and the printed equation numbers. `figures.py` finds the figures and writes crop-check images.
2. Claude views the strips and writes each page as Markdown to its own file, zooming to 500–600 dpi wherever a bar, tilde, prime, typeface or brace extent is in doubt.
3. Writing conventions: printed equation numbers in `\tag{}`; typefaces exactly as printed ($\mathcal{K}$, K, $\mathbf{\Omega}$, upright i, d, e where so printed); a placeholder for each figure followed by its transcribed caption; paragraphs joined across page breaks.
4. `assemble.py` joins the pages, writes the two output files, and `check.py` tests every formula in KaTeX, the equation numbers against the PDF, and the absence of text-layer debris.
5. `verify_eqs.py` re-typesets each numbered equation under the PDF's original for a symbol-by-symbol proofread.
6. Claude delivers `NAME.md` and, if there are figures, `NAME_figures.pdf`.

File	Role
`SKILL.md`	Instructions: ground rules, procedure, conventions, budget.
`scripts/prepare.py`	Strips, a searchable text layer, math-font share and equation numbers per page; one printed line per page.
`scripts/figures.py`	Finds each figure from its caption, renders the whole region at 200 dpi, writes a check image with the crop outlined.
`scripts/assemble.py`	Joins the pages; writes `NAME.md`; packs the figures into `NAME_figures.pdf` and writes the page reference for each.
`scripts/check.py`	KaTeX syntax, equation numbers, debris.
`scripts/verify_eqs.py`	Side-by-side proofreading sheets.
`scripts/eval/`	`mathcmp.py` and `prosecmp.py`: scoring against a LaTeX source (testing only).

Output naming. `NAME` is the original PDF's file name without `.pdf`, unchanged: same characters, case, spaces and underscores. In `NAME.md` each figure appears as a line such as `*[Fig. 4: NAME_figures.pdf, page 1]*` followed by its caption, so Claude knows where to look.

2 Why Markdown instead of the original PDF

All numbers in this section are estimates for the eight-page, two-column PNAS paper used in testing (Lihm, Kiese, Lee & Kugler, PNAS **123**, e2525308123, 2026), based on Anthropic's published approximation for image tokens (width × height /750 after scaling to at most 1,568 px and about 1.15 megapixels) and on character counts for text. None is a metered measurement.

2.1 What Claude receives

From a PDF, Claude receives the extracted text and, for each page, an image. The text layer garbles mathematics: calligraphic and roman letters merge, bold is lost, accents detach (in testing, $\tilde{\mathrm{K}} = 1/\Omega_{\bar{1}}$ came out as `K = 1 ~ /Ω¯1`), and big operators become placeholder codes. A full page squeezed into one image is seen at only about 111–114 dpi, close to where subscripts, bars and primes became uncertain in the legibility test (Section 4). From the Markdown, Claude receives every equation as exact LaTeX text, with no reading involved.

2.2 Project capacity

Uploaded per paper	Tokens	Papers in ∼200K
Original PDF (text layer ∼15–19K + 8 page images ∼12K)	27,000–31,000	5–6
`NAME.md` + `NAME_figures.pdf` (3 pages, ∼3,600)	19,000–24,000	7–9
`NAME.md` only	15,000–20,000	8–11
`NAME.md` without the reference list	10,000–14,000	12–16

The text costs about the same in both forms (58,000 characters for this paper); the saving comes from replacing eight full-page images by three pages of figures, or by none. With all figures included the Markdown route uses about 20–30% less capacity; without the figures PDF, about 40–45% less. Leaving out the reference list, a third of this paper's text, saves more than any choice about figures. Project knowledge is read into every conversation of the Project, so the same percentages apply to the standing cost of each chat.

When a Project's knowledge approaches the context limit, Claude switches to retrieval and searches the files instead of reading them whole. Clean LaTeX and text are found by such a search; the content of page images is not.

2.3 Working with equations in a chat

Reading an equation reliably from a PDF means the careful route of the academic-pdf-reading skill: its instructions (measured: about 5,200 tokens), a triage of the file, locating the equation, and a cropped, enlarged image of it for every equation, followed by a transcription.

Extra cost	Image route	Markdown
Setup per chat (instructions, triage)	∼6,000–7,000	0
Per equation question, new tokens	∼1,500–4,000	∼0
Per equation question, tool round trips	4–7	0

The crop images themselves are cheap (about 100–250 tokens); the cost is the work around them. Two scenarios:

Scenario	Extra new tokens	Round trips	Cached re-reads
10 equation questions in one chat	21,000–47,000	40–70	1.5–2.5 M
10 equations referred to 10 times each, each read once and reused	21,000–47,000	40–70	1.5–2.5 M
the same, each re-read from the image every time	95,000–270,000	220–340	9–14 M
Markdown in the Project, any of the above	∼0	0	0

Reusing an equation once read is what a careful run does, but long chats are summarised automatically near the context limit, which can discard the crops and force re-reading. Every round trip re-reads the conversation; cached reads count less towards usage limits than new tokens, but not nothing. In an equation-heavy chat the saving is therefore three to six times the standing capacity saving of a paper, and it grows with the length of the chat.

2.4 Accuracy

The Markdown is verified once, at conversion, with checks and proofreading. Reading from the PDF repeats the risk of a misreading in every chat, and a misread equation that is reused carries the error into every later reference.

2.5 Break-even

A conversion costs about 4,000–5,500 tokens per page, about 35,000–45,000 for an eight-page paper, once. That is roughly one equation-heavy chat on the image route, so the conversion pays for itself within one or two such chats and keeps paying afterwards.

2.6 Caveats

- Whether PDFs in Project knowledge are given page images, as PDFs attached in a chat are, has not been verified. If Projects use only the extracted text, the original PDF takes about as much capacity as the Markdown alone: no saving in capacity, but garbled equations and invisible figures remain.
- Chats that are mostly conceptual and rarely touch equations gain little beyond the capacity saving.
- Two-column papers such as the PNAS test paper cannot yet be converted reliably as text (Section 7); for them the comparison above becomes available once column support is built.
- Before relying on the numbers, compare the usage panel before and after one chat of each kind.

2.7 Recommendations

Upload `NAME.md` for every paper. Add `NAME_figures.pdf` only for papers whose figures you will discuss; the captions are always in the text. Consider removing the reference list from papers whose references you do not need Claude to read.

3 Figures and adaptive packing

Detection. `figures.py` anchors each figure on its caption (a text block starting "Fig. N", "Figure N" or "FIG. N"). The figure is everything graphical above the caption within its column, up to the nearest running text. The region is rendered whole at 200 dpi, because physics figures are usually composites of vector art and small embedded images such as heatmaps, which extraction would split into fragments. Decorative graphics without a caption are ignored. A check image with the crop outlined in red is viewed for every figure.

Packing. Claude sees a PDF page as one image of at most about 1.15 megapixels, roughly 1,500 tokens. More figures per page cost fewer tokens each but get fewer pixels. `assemble.py` places figures at native size in order of appearance and starts a new page when the next figure would push the page below 150 dpi effective: an area of about 50 square inches or a long edge of about 10 inches. 150 dpi is the level at which small labels stayed reliably legible in testing. A large figure shares its page with at most one small one; a figure too large for any page goes alone and is reported. Each figure is labelled "Fig. N", and the Markdown names its page.

Fixed figures per page (model, PNAS figures)	Pages	Effective dpi	Tokens
2	4	190–205	≤5,800
3	3	150–165	≤4,600
4	2	150–155	≤3,100
6	2	∼120	≤3,100
7	1	∼110	≤1,500
Adaptive rule (actual output)	3	152, 183, 200	∼3,600

The adaptive rule packed the seven figures 4, 2 and 1 per page: adding the seventh to page 2 would have taken that page below the floor. Bundling does not reduce tokens below separate images of equal legibility (the seven figures as 200 dpi PNGs: about 3,300 tokens); its value is a single file that a Project accepts.

4 Test results

Documents. Kugler note (five pages, single column, with LaTeX source); duality note (ten pages, single column, with source); PNAS paper (eight pages, two columns, no source). All transcriptions were made from images before the source was opened.

Test	Result
Kugler note, whole document	25/26 displays, 143/147 inline exact; prose identical
Kugler note, page 3 (densest notation)	8/8 displays, 38/38 inline exact
Duality note, pages 4 and 6, open blind at 170 dpi	10/10 displays, 70/70 inline, strict
Blind legibility, 16 items × 6 resolutions (232–100 dpi)	96/96 correct (forced choice)
Proofreading, 9 planted errors	6 caught on sheets, 3 by compile check, 0 missed
Scorer, 4 planted errors	4/4 flagged; spelling variants forgiven
Figures (PNAS): found, crops complete, decorations ignored	7/7, 7/7, 0 false
Figure captions (PNAS) in KaTeX	48/48
Figures PDF, most crowded page viewed at ∼151 dpi	all labels legible
Output names, including `My Paper (2026) v2.PDF`	exact; references correct

The five whole-document mismatches in the Kugler note render identically to the source (spelling-only differences such as `F_{\rm rest}` versus `F_{\mathrm{rest}}`). In the legibility test, about five readings felt uncertain at 120 and 100 dpi and none at 145 dpi or above apart from one overbar extent. On the most crowded figures page, colorbar exponents, inset tick labels, the superscripts of $i_{\mathrm{D}}^{\mathrm{DMFT}}$ and $\tilde{I}_{\mathrm{D}}^{\mathbf{q}=\Gamma}$ and the legend subscripts $\mathbf{k}_{\mathrm{N}}$, $\mathbf{k}_{\mathrm{HS}}$, $\mathbf{k}_{\mathrm{AN}}$ were all read without zooming. Caption accuracy of the PNAS paper was not scored, because its source was not available. Marker could not be compared: its model weights are hosted on sites Claude's sandbox cannot reach; a page-3 comparison kit is provided for a local run.

5 Cost of a conversion

Per full page: strips about 2,400 tokens, Markdown output about 1,200, proofreading about 380 per numbered equation, zooms 0–500; in total about 4,000–5,500, plus about 1,400 for `SKILL.md` once per conversion. Every strip stays in the conversation, so a ten-page note leaves about 40,000–55,000 tokens in context: convert one paper or chapter per conversation, in a fresh chat. Against a Pro plan's five-hour window, community estimates put this at very roughly 1–4% per page with Opus; Anthropic publishes no token figure for the limit.

6 Changes considered and not made

- **Selective proofreading (rejected).** Errors were found in the plainest equations, and only 1 of 26 equations in the Kugler note would have been skippable.

- **Skipping strips without math fonts (deferred).** Saves 0–6% of strip tokens, and first needs scan detection and hyphen repair.
- **Images embedded in the Markdown (removed).** Claude cannot see a base64 image in a text file, and it costs an estimated 100 times the tokens of the same image as a file.

7 Known limitations

- **Scanned PDFs (serious).** A scan has no fonts, so `prepare.py` marks every page `text-ok`; scanned pages must be transcribed from images regardless.
- **Two-column layouts.** Strips run across both columns; equation numbers are found only in the right quarter of the page; proofreading crops include the other column. Figures are not affected.
- **Figures.** Found only when a caption below them starts with "Fig.", "Figure" or "FIG."; other cases need a manual crop. Text inside figures is part of the image. In a Project, Claude sees figures only if the figures PDF is uploaded.
- **Dotted equation numbers** such as (5.40) are not recognised, so `check.py` wrongly fails on them.
- **Proofreading sheets** omit equations that do not compile, merge adjacent `align` lines, and can clip large symbols.
- **Fonts.** Math detection relies on font names; unusual or bitmap math fonts may be misclassified.
- **Self-proofreading.** Without a source, the same model transcribes and proofreads; sheets cover numbered displays only.
- **Unverified assumptions.** How Projects process PDF pages, and all token figures, are modelled rather than measured.

8 Installation and requirements

Upload `pdf-to-markdown.zip` under *Customize* → *Skills* with code execution enabled, replacing any earlier version. The scripts need PyMuPDF and Pillow; `verify_eqs.py` needs pdflatex; `check.py` needs Node.js and installs KaTeX itself. All were present in Claude's sandbox during testing.

9 Planned next

Column support for strips, equation numbers and proofreading crops, followed by a full two-column conversion; scan detection and a scanned-page test; dotted equation numbers; placeholder panels and better crops on the proofreading sheets; other caption placements for figures; a metered comparison of Markdown and original PDF in a real Project; the Marker comparison, run locally.